AAE 54600 – Aerospace Structural Dynamics and Stability
3 credits

1 Course Information/Description

This course will examine structural dynamics in terms of mechanics, kinematics, vibrations, spectral/modal analysis, wave propagation, and stability of motion. This course will also use MATLAB for homework wherever necessary to increase the learning outcome of the course.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:

Recommended but not necessary: Nonlinear structural dynamics using FE methods by Prof. J. F. Doyle, Cambridge University Press, 2015.

3 Assignments

Weekly homework will be posted on Brightspace.

4 Course Learning Objectives

On completing this course, students should be able to perform dynamic analysis of given engineering problems through:

- Modal analysis
- Spectral analysis
- Wave propagation
- Concept of Stability

5 Evaluation

Assignments and Exams	Due	Percentage
Homework	Weekly	50%
Exam 1	03/05/2026	25%
Exam 2	04/30/2025	25%
		Total: 100%

- You are allowed one homework extension without incurring a late submission penalty.
    - You may use this extension on any homework assignment.
    - You must notify the instructor of your intent to use your extension at least 24 hours in advance of the original submission deadline.
    - Extension requests received after the original deadline has passed will not be entertained.
    - The maximum extension that will be granted is 48 hours after the original submission deadline.
    - Once you have used your extension, further requests for extensions will not be entertained.
    - Exceptions will be made for legitimate medical emergencies where documentation can be produced.
- Any homework submitted late (without an approved extension) will receive a zero.
- Homework regrade requests will be accepted for one week after homework and exam grades are published.
- There will be two mid-term exams. Each exam is worth 25% of your grade. Reasonable accommodation will be made for students having exceptional circumstances preventing them from taking exams during the scheduled times. However, requests for such accommodation must be made as soon as possible. The instructor reserves the right to use his discretion regarding this, and conflicts will be mediated through the college/university.

Use of artificial intelligence (AI) or Large Language Models (LLM):

- Any use of Artificial Intelligence or Large Language Models (AI/LLM) to generate solutions or parts of a solution to the homework or exams will be considered a blatant attempt at cheating and will result in a failing grade in the course.

Your grade will be calculated as follows:

- Accident Case Study: 25%
- Weekly Reflections: 20%
- Team Term Project: 50%
- Final Project Peer Review: 5%

Your course grade will be based on the following grading scale:

Letter Grade	Percentage
A+	97-100%
A	92-96%
A-	90-91%
B+	87-89%
B	84-86%

Letter Grade	Percentage
D+	67-69%
D	64-66%
D-	60-63%
F	0-59%

B-	80-83%
C+	77-79%
C	74-76%
C-	70-73%